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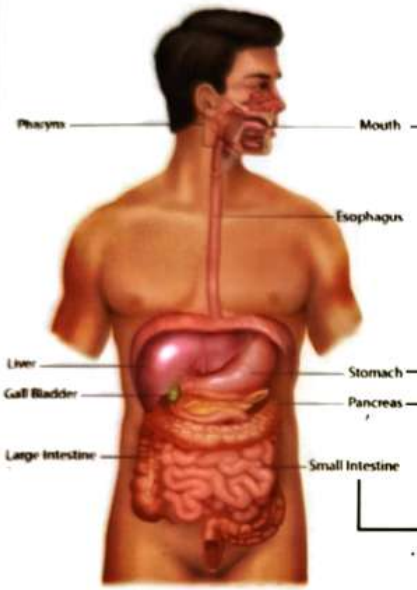
DIGESTION AND ABSORPTION

----- Active space --

Carbohydrates

00:01:05

DIGESTION : **Diet : Polysaccharides.**



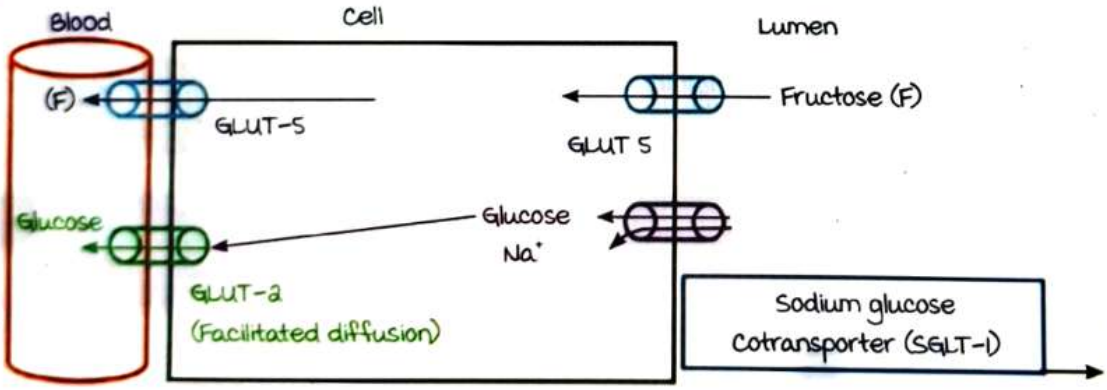
Mouth → Salivary amylase (In saliva)
- Activated by chloride ions.
- Inhibited by gastric acid.

Stomach → No enzymes play a role in digestion of carbohydrates.

Pancreas → Pancreatic amylase :
- Major enzyme.
- Acts on 1:4 α linkages → Disaccharides.

Disaccharidases (Brush border enzyme).
- Eg : Lactase, isomaltase, sucrase.
- Acts on 1:6 α linkages → monosaccharides (Glucose, Fructose).

ABSORPTION :



- Found in GIT.
- Secondary active transport.

Clinical aspect :

- ORS : Na+ absorption is facilitated by glucose absorption (via SGLT-1).
  1. Congenital glucose galactose malabsorption (CGGM).
     - Congenital absence of SGLT-1.
     - Genetics : Autosomal recessive.
     - Presentation : Chronic refractory diarrhoea.

Note : SGLT-2 is found in Kidney.

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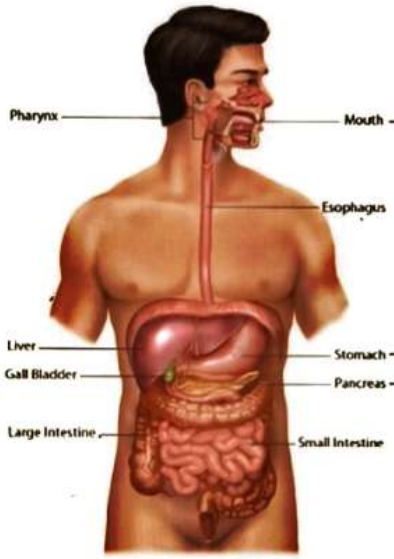
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Lipids

00:12:20

DIGESTION :

Diet : Triglycerides (TG)



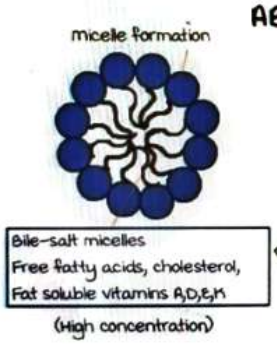
Mouth → Lingual lipase

Stomach → Gastric lipase

Pancreas → Pancreatic lipase
Colipase
Cholesterol esterase

} Act on TG and forms free fatty acids.

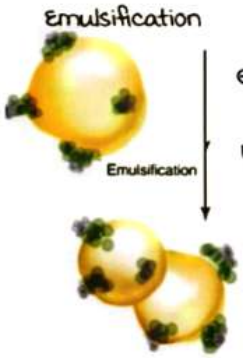
ABSORPTION



Passive Diffusion

Mucosal cell (Low concentration)
Free fatty acids
↓ Reesterification
Triglycerides
↓ Transported by
Chylomicrons

Emulsification

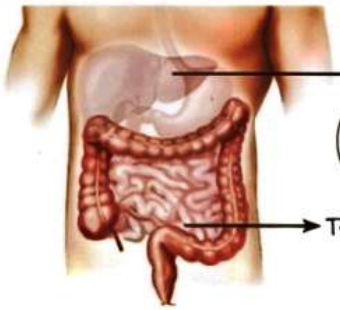


mechanism : Bile salt + fat.

Globule → Prevent formation of large fat globule.

Use : ↑ Action of Lipase (Facilitates digestion & absorption of lipids).

Enterohepatic circulation :



Reduce forming new bile salt : Conserve energy.

Note :

Short chain fatty acids (SCFAs) :

- Eg : Propionate, butyrate, acetate.
- Formed by colonic flora.
- Functions :
  - Trophic action : ↑ mucosal growth.
  - Acid base balance : Absorbed in exchange for H^+.
  - Prevents inflammation.

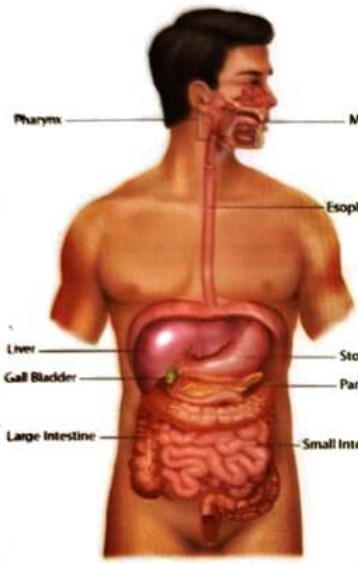
Feedback

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Protein

00:23:15 ----- Active space -----

DIGESTION :

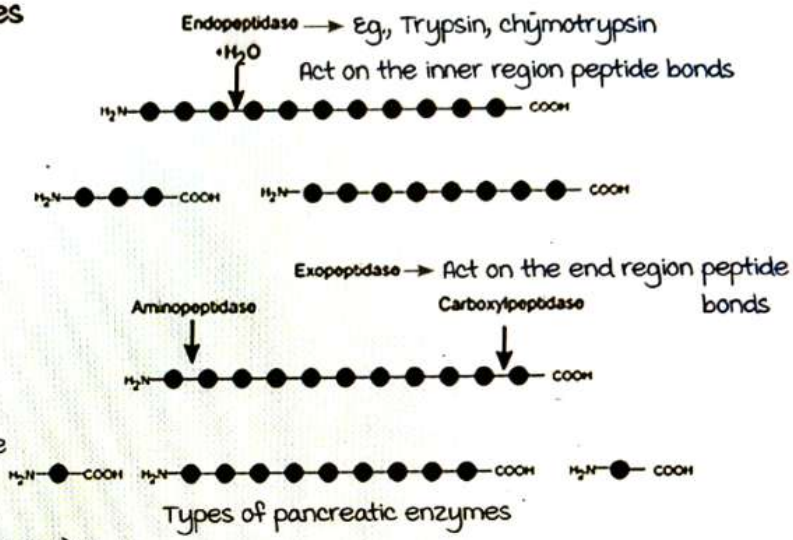


Diet : Polypeptides

Mouth → No enzymes for protein digestion

Stomach → Pepsin

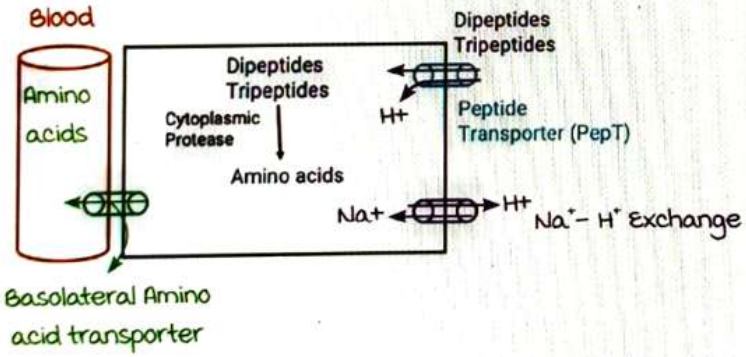
Pancreas → Trypsin, Chymotrypsin, Carboxypeptidase



Types of pancreatic enzymes

- Present in form of zymogen (Inactive).
  - Prevent autodigestion of pancreas.
- Activated in small Intestine.
- Form dipeptides & tripeptides.

ABSORPTION :



Vitamins, Water absorption

00:29:35

VITAMIN ABSORPTION :

Fat soluble vitamins : Absorbed along with micelles.

Water soluble vitamins (B complex) : Absorbed by sodium dependent cotransporters.

Except : Vitamin B12 & Folate : Transport independent of Na+.

WATER ABSORPTION :

- Total water content in GI tract : 9L.
- Water reabsorbed : 8.8L.
- Water excreted in stools : 200mL.
- Maximum water absorbed in Jejunum.

Feedback

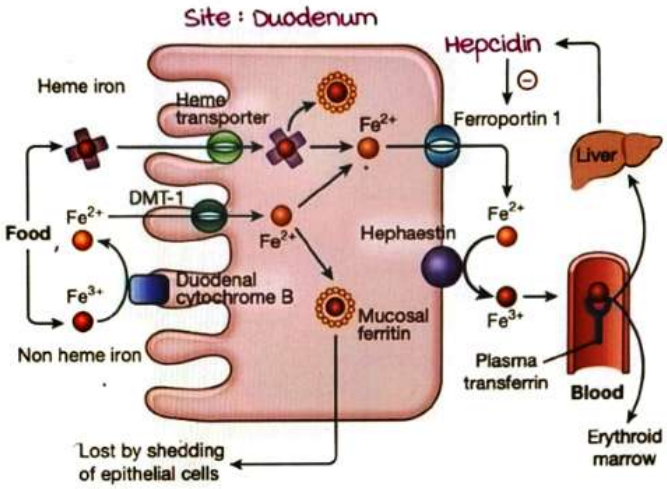
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Iron Absorption

00:32:26

MECHANISM :



Iron absorption depends on :

1. Dietary iron intake.
2. Body iron stores.
3. State of erythropoietic activity (In bone marrow).

Factors ↑
Vitamin C

Factors ↓
Tannins
Oxalates
Phytates

REGULATION :

Hepcidin (Negative regulator) :

Hepcidin → released from liver.

Genetics : HFE gene $\xrightarrow{\text{Mutation}}$ Hemochromatosis.

Function : Represses basolateral transport of iron into blood.

Binds to ferroportin → Iron not released from macrophages & other cells.

Links body iron stores with iron absorption.

Feedback